Project Proposal (Group 07)

1. Team Members and TA Consulted

Members: Aditya Tripuraneni (tripuraa), Harmanpreet Singh Sagar (sagarh), and Vanessa Lai (laiv3)

TA Consulted and Date: Yasmin - September 28, 2026

2. Task Title and Overview

Task Title: Log-Based Anomaly Detection in Distributed Systems: A Supervised vs. Unsupervised Comparison on HDFS

Overview: System logs are the primary record of what a distributed system did, but modern systems produce millions of log lines per day, far too many for manual review. Our project detects anomalous program executions in HDFS logs by comparing supervised classifiers trained on labelled sessions against unsupervised methods trained without labels to understand how much the labels are actually worth. This task is challenging because free-text logs must be parsed into structured event templates before any model can use them, anomalies are rare enough that naive models score well while missing nearly every failure, and the order of events within a session carries information that a bag-of-events representation discards.

3. Task Definition

Type of Task: Binary Sequence Classification

Input: One HDFS block session, represented as an ordered sequence of parsed log event-template IDs.

Output: Single label per session, either normal or anomalous

Number of Classes: 2, single-label

4. Problem Description, Real World Impact, and Challenge

Engineers frequently spend hours searching through logs when a production system fails to determine which operation went wrong and why. This manual process doesn't scale as systems grow, is slow, and depends on individual expertise. Automated log anomaly detection reduces the time needed to detect and diagnose errors, lowering downtime and its associated costs. Platforms like Datadog apply machine learning to logs for this purpose, which reflects real industry demand. Producing labels also requires domain expertise, which is why comparing supervised accuracy against label-free methods has real practical value.

5. Data Sources and Plan for Data Collection

Data Source: We'll be using the **HDFS_v1** dataset from **Loghub**, which is a curated collection of system log datasets managed by the LogPAI research group for AI-driven log analytics research

(Loghub and HDFS Dataset). The dataset is publicly available for research purposes. Since it is a direct download, rate-limit management, API access, and scraping are not required.

How the data was collected: HDFS_v1 was originally produced by Xu et al. [1] by running benchmark workloads on a 203-node HDFS cluster in a private cloud environment, yielding 1.47 GB worth of data over 38.7 hours [2].

How the data was labelled: Logs are sliced into traces by HDFS block ID (blk_…), giving one session per block. Each session was manually labelled normal or anomalous using handcrafted rules. No hand-labelling is required on our part.

Metadata available: Each raw line contains a date, time, process ID, log level, component, and message content. Loghub also provides the following preprocessed files: HDFS_templates.csv, anomaly_label.csv, Event_traces.csv, Event_occurrence_matrix.csv, and HDFS.npz.

Pre-processing plan: Parse raw messages into event templates using Drain. Then group events by block ID into ordered sequences. Finally, join the sequence with the respective label.

Amount of Dataset Utilized: The entire HDFS_v1 dataset will be utilized. If training the sequence model proves to be too expensive, per the TA recommendation, we will employ a stratified random subset maintaining the initial anomaly rate.

6. Expected Size of Dataset

The dataset contains **575,061 labelled block sessions**, of which **16,838 (2.93%)** of them are anomalous. Each data point is one block session. The following table showcases three sample labelled data points:

blk_7503483334202473044 → [E5, E5, E22, E5, E11, E9, E11, E9, E11, E9, E26, E26, E26, E3, E2, E2] → Normal

blk_-9073992586687739851 → [E5, E22, E5, E5, E11, E9, E11, E9, E11, E9, E26, E26, E26, E2, E2, E2] → Normal

blk_-3102267849859399193 → [E5, E5, E22, E7, E11, E10, E14, E7] → Anomalous

The raw files and the event sequence are available at the following locations: HDFS dataset

The event sequence definitions are available here: HDFS_templates.csv

7. Proposed Solution

Approach: We will parse the raw logs into event templates using Drain and group them into one sequence per block [3]. All models will then use the same preprocessing and the same stratified 70/10/20 train/validation/test split. The unsupervised models will never see labels during training; labels are used only for evaluation.

Features and labels: Each session is represented in the following two ways:

- Event-count vectors, with TF-IDF and n-gram variants. The n-grams keep some event order.
- The raw ordered event sequence, used by the LSTM.

The target is a binary Normal/Anomalous label for each session.

Models: Supervised models would involve logistic regression as an interpretable baseline and XGBoost [1]. Both use class weighting to handle the imbalance. For unsupervised models, Isolation Forest [5] on the count vectors and a DeepLog-style LSTM [6] will be utilized. The LSTM flags a session when its events fall outside the model's top-k next-event predictions, which captures the sequential structure the TA pointed out. If the LSTM proves too costly, we will substitute a PCA-based detector [1]. To measure how much labels are worth, we will also retrain the supervised models on 1%, 10%, and 100% of the labels.

Existing solutions: HDFS is a standard benchmark for log anomaly detection. Prior work reports results for PCA, invariant mining, logistic regression, SVM, and deep sequence models such as DeepLog [3, 7]. Our contribution is a controlled, same-pipeline comparison of supervised and unsupervised methods, rather than a new model.

Evaluation: We will report precision, recall, F1, and PR-AUC on the anomalous class. We will not use accuracy, because predicting "Normal" for every session already scores about 97%. Unsupervised decision thresholds will be tuned on the validation set only. We will judge a model as good if it is competitive with the published HDFS benchmarks.

Libraries: Python, pandas, NumPy, scikit-learn, XGBoost, PyTorch, Drain3, and Matplotlib.

8. TA Discussion

Our team met with Yasmin, our TA, on September 28, 2026, and discussed numerous items. We asked how many models would count as sufficient. She advised that two models per family (supervised and unsupervised) is an appropriate scope, so we kept our plan at four models. We asked whether framing the project as a supervised-versus-unsupervised comparison was acceptable. She confirmed that it was. Since the course does not allow simple regression models, we asked whether logistic regression (a classifier) was allowed. The TA confirmed it is acceptable when paired with stronger models, so we use it as an interpretable baseline alongside XGBoost. She also suggested we consider whether our data is sequential or time-series in nature and use models suited to that. This led us to include the DeepLog-style LSTM and n-gram features that keep event order. Finally, the TA suggested using a subset if the full dataset proves too large, so we added the stratified-subset fallback in Section 5.

9. References

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[3] S. He, J. Zhu, P. He, and M. R. Lyu, 'Experience Report: System Log Analysis for Anomaly Detection', in *2016 IEEE 27th International Symposium on Software Reliability Engineering (ISSRE)*, 2016, pp. 207–218. doi: 10.1109/ISSRE.2016.21.

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